Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. We trained each model for 100 epochs with a batch size of 32.

2 Results

Figure 1 shows the accuracy during training.

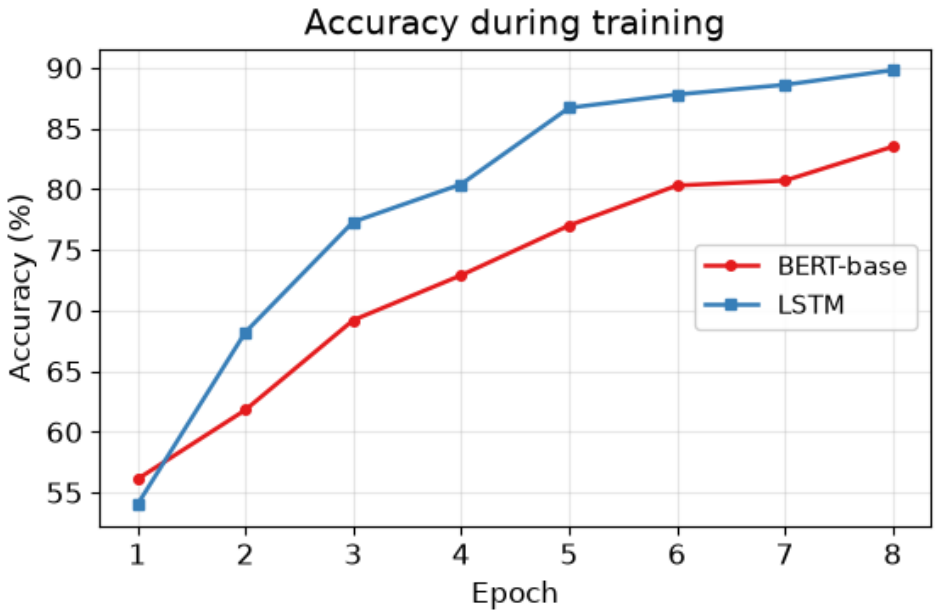


Figure 1: Accuracy during training for the models in this experiment.

From epoch 7 to epoch 8, the accuracy of LSTM improves by a relative 0.5% (Figure 1). At epoch 2, BERT-base achieves 62% accuracy.

Figure 2 shows the f1 score of EfficientNet by dataset.

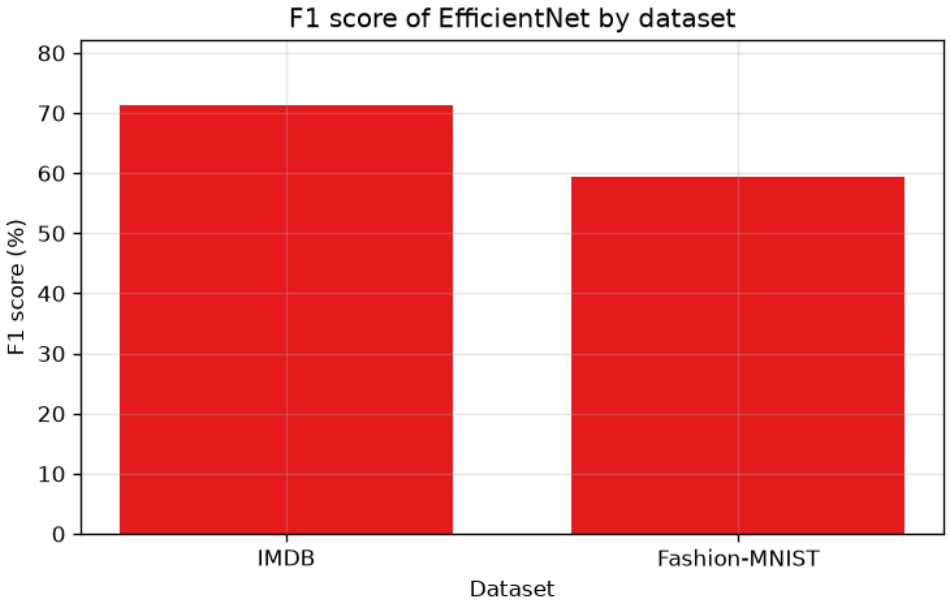


Figure 2: F1 score of EfficientNet by dataset for the models in this experiment.

EfficientNet scores 16.2 points higher on IMDB than on Fashion-MNIST. On Fashion-MNIST, EfficientNet obtains a F1 score of 100.4%. EfficientNet scores 20.3 points higher on IMDB than on Fashion-MNIST. On

Fashion-MNIST, EfficientNet obtains a F1 score of 59% (Figure 2).

Figure 3 shows the f1 score by dataset.

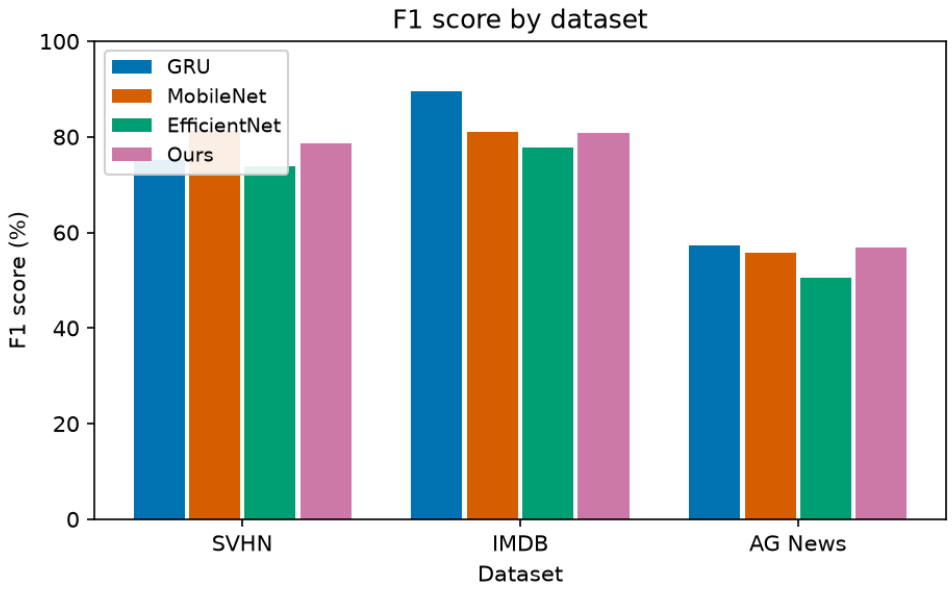


Figure 3: F1 score by dataset for the models in this experiment.

EfficientNet reaches about 80% F1 score on IMDB. On SVHN, Ours is 3.6 points better than GRU. As shown in Figure 3, Ours achieves 66.8% F1 score on AG News.

Figure 4 shows the f1 score by dataset.

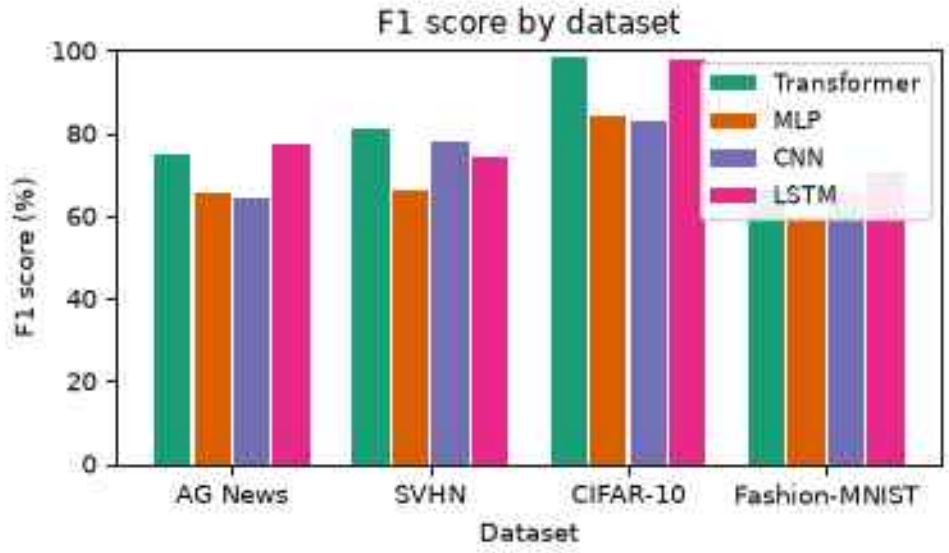


Figure 4: F1 score by dataset for the models in this experiment.

As shown in Figure 4, MLP achieves 59.9% F1 score on Fashion-MNIST. On SVHN, CNN obtains a F1 score of 78%. CNN achieves 83% F1 score on CIFAR-10. As shown in Figure 4, CNN achieves 78.2% F1 score on SVHN.

3 Discussion

Logistic Regression reaches 51.3% accuracy on SVHN. We note that the training was stable for all models.